

# Black Box testing bug report

Roll Number: 2024111006

## [BUG-001] Authentication Bypass: Negative Roll Number

- **Endpoint:** GET /api/v1/admin/users
- **Request:** GET | Headers: {"X-Roll-Number": "-1"}
- **Expected:** 401 Unauthorized
- **Actual:** 200 OK

## [BUG-002] Authentication Bypass: Zero Roll Number

- **Endpoint:** GET /api/v1/admin/users
- **Request:** GET | Headers: {"X-Roll-Number": "0"}
- **Expected:** 401 Unauthorized
- **Actual:** 200 OK

## [BUG-003] Header Validation Inconsistency: Roll Number Length

- **Endpoint:** GET /api/v1/admin/users
- **Request:** GET | Headers: {"X-Roll-Number": "1" \* 20}
- **Expected:** 401 Unauthorized (per strict roll length expectations)
- **Actual:** 400 Bad Request

## [BUG-004] Path Extraction Failure: Negative Admin User ID

- **Endpoint:** GET /api/v1/admin/users/-1
- **Request:** GET
- **Expected:** 400 Bad Request
- **Actual:** 404 Not Found

## [BUG-005] Profile Data Type Bypass: Alphabetic Phone Number

- **Endpoint:** PUT /api/v1/profile
- **Request:** PUT | Body: {"phone": "ABCDEFGH IJ"}
- **Expected:** 400 Bad Request
- **Actual:** 200 OK

## [BUG-006] Profile Data Type Bypass: Hyphenated Phone Number

- **Endpoint:** PUT /api/v1/profile
- **Request:** PUT | Body: {"phone": "123-456-78"}
- **Expected:** 400 Bad Request
- **Actual:** 200 OK

## [BUG-007] Address Data Type Bypass: Alphabetic Pincode

- **Endpoint:** POST /api/v1/addresses
- **Request:** POST | Body: {"pincode": "ABCDEF"}
- **Expected:** 400 Bad Request
- **Actual:** 200 OK

## [BUG-008] Address Modification Rule Violation: Label Update

- **Endpoint:** PUT /api/v1/addresses/1
- **Request:** PUT | Body: {"label": "OFFICE"}
- **Expected:** 400 Bad Request (Label is immutable via PUT)
- **Actual:** 200 OK

## [BUG-009] Address Schema Violation: Non-Boolean Default Status

- **Endpoint:** PUT /api/v1/addresses/1
- **Request:** PUT | Body: {"is\_default": "X"}
- **Expected:** 400 Bad Request
- **Actual:** 200 OK

## [BUG-010] Search Logic Failure: No matching results

- **Endpoint:** GET /api/v1/products?name=ZZZZZZ
- **Request:** GET
- **Expected:** [] (Empty List)
- **Actual:** Full catalog (210 items)

## [BUG-011] Sort Logic Failure: Price Descending

- **Endpoint:** GET /api/v1/products?sort=price&order=desc
- **Request:** GET
- **Expected:** [Price: 250, Price: 220, ...]
- **Actual:** [Price: 120, Price: 130, ...]

## [BUG-012] Path Extraction Failure: Negative Product ID

- **Endpoint:** GET /api/v1/products/-1
- **Request:** GET
- **Expected:** 400 Bad Request
- **Actual:** 404 Not Found

## [BUG-013] Path Extraction Failure: Alphabetic Product ID

- **Endpoint:** GET /api/v1/products/ABC
- **Request:** GET
- **Expected:** 400 Bad Request
- **Actual:** 404 Not Found

## [BUG-014] Cart Validation Bypass: Zero Quantity

- **Endpoint:** `POST /api/v1/cart/add`
- **Request:** `POST` | Body: `{"product_id": 1, "quantity": 0}`
- **Expected:** `400 Bad Request`
- **Actual:** `200 OK`

## [BUG-015] Cart Math Violation: Negative Subtotals

- **Endpoint:** `GET /api/v1/cart`
- **Request:** `GET`
- **Expected:** `Subtotal ≥ 0`
- **Actual:** `subtotal: -16` (Observed for ID 1)

## [BUG-016] Cart Aggregation Violation: Logical Total Sum

- **Endpoint:** `GET /api/v1/cart`
- **Request:** `GET`
- **Expected:** `total == Sum(subtotals)`
- **Actual:** `120 != -6` (Mismatch in aggregation)

## [BUG-017] Cart State Vacuum: Total Zero for Non-Empty

- **Endpoint:** `GET /api/v1/cart`
- **Request:** `GET`
- **Expected:** `total > 0` (for non-empty cart)
- **Actual:** `total: 0`

## [BUG-018] Cart Update Logic Failure: Non-Existent Item

- **Endpoint:** `POST /api/v1/cart/update`
- **Request:** `POST` | Body: `{"product_id": 2, "quantity": 5}` (ID 2 not in cart)

- **Expected:** 404 Not Found
- **Actual:** 200 OK

## [BUG-019] Cart Schema Failure: Missing ID Parameter Result

- **Endpoint:** POST /api/v1/cart/add
- **Request:** POST | Body: {"quantity": 1}
- **Expected:** 400 Bad Request
- **Actual:** 404 Not Found

## [BUG-020] Cart Schema Failure: Accept Missing Quantity

- **Endpoint:** POST /api/v1/cart/add
- **Request:** POST | Body: {"product\_id": 1}
- **Expected:** 400 Bad Request
- **Actual:** 200 OK

## [BUG-021] Coupon Handling Inconsistency: Fake Coupon Status

- **Endpoint:** POST /api/v1/coupon/apply
- **Request:** POST | Body: {"coupon\_code": "FAKE"}
- **Expected:** 404 Not Found
- **Actual:** 400 Bad Request

## [BUG-022] Coupon Handling Inconsistency: Case Sensitivity Status

- **Endpoint:** POST /api/v1/coupon/apply
- **Request:** POST | Body: {"coupon\_code": "mega15"}
- **Expected:** 404 Not Found
- **Actual:** 400 Bad Request

## [BUG-023] Checkout Bypass: Empty Cart Order

- **Endpoint:** POST /api/v1/checkout
- **Request:** POST | Body: {"payment\_method": "CARD"} (Empty Cart)
- **Expected:** 400 Bad Request
- **Actual:** 200 OK

## [BUG-024] Checkout Rule Violation: COD Threshold Bypass

- **Endpoint:** POST /api/v1/checkout
- **Request:** POST | Body: {"payment\_method": "COD"} (Total > 5000)
- **Expected:** 400 Bad Request
- **Actual:** 200 OK

## [BUG-025] Wallet Validation Bypass: Floating Point Value

- **Endpoint:** POST /api/v1/wallet/add
- **Request:** POST | Body: {"amount": 1.5}
- **Expected:** 400 Bad Request (Monetary values must be integer)
- **Actual:** 200 OK

## [BUG-026] Admin Schema Inconsistency: Loyalty Points Key Mismatch

- **Endpoint:** GET /admin/users/1
- **Results:** loyalty\_points: 448 vs /loyalty result None (Key mismatch or points missing).

## [BUG-027] Path Extraction Failure: Alphabetic Order ID

- **Endpoint:** GET /api/v1/orders/ABC
- **Expected:** 400 Bad Request
- **Actual:** 404 Not Found

## [BUG-028] Products Integrity Violation: Non-Existent Product Reviews

- **Endpoint:** `GET /api/v1/products/999/reviews`
- **Expected:** `404 Not Found`
- **Actual:** `200 OK` (returns empty list for non-existent root)

## [BUG-029] Products Schema Failure: Missing Average Rating

- **Endpoint:** `GET /api/v1/products/2`
- **Expected:** JSON key `average_rating` (implied by documentation)
- **Actual:** Key completely missing from response.

## [BUG-030] Support Validation Bypass: Subject Length (V.Short)

- **Endpoint:** `POST /api/v1/support/ticket`
- **Request:** `POST` | Body: `{"subject": "A"}`
- **Expected:** `400 Bad Request` (Min length 5)
- **Actual:** `200 OK`

## [BUG-031] Ticket Lifecycle Violation: CLOSED -> OPEN Rollback

- **Endpoint:** `PUT /api/v1/support/tickets/{id}`
- **Request:** `PUT` | Body: `{"status": "OPEN"}` (Existing status is CLOSED)
- **Expected:** `400 Bad Request`
- **Actual:** `200 OK`

## [BUG-032] Ticket Schema Violation: Arbitrary Status Acceptance

- **Endpoint:** `PUT /api/v1/support/tickets/{id}`

- **Request:** PUT | Body: {"status": "FAKE"}
- **Expected:** 400 Bad Request
- **Actual:** 200 OK

## [BUG-033] Authentication Bypass: Roll Number Padding

- **Endpoint:** GET /api/v1/admin/users
- **Request:** GET | Headers: {"X-Roll-Number": " 2024"}
- **Note:** This bypasses standard integer sanitation in the request layer.